

1) Rotation matrix $M(\theta, \phi)$

such that $\vec{a} = M(\theta, \phi) \cdot \vec{a}'$

we have $\hat{n} = (\sin\theta \cos\phi, \sin\theta \sin\phi, \cos\theta)$

with $\vec{a} = R\hat{a}$

we want $\hat{z}' = \hat{n}$

$$= (\sin\theta \cos\phi, \sin\theta \sin\phi, \cos\theta)$$

$\hat{z}' \cdot \hat{x}' = 0$ allows

$$\hat{x}' = (\cos\theta \cos\phi, \cos\theta \sin\phi, -\sin\theta)$$

$$\vee \hat{z}' \times \hat{x}' = \hat{y}' = (-\sin\phi, \cos\phi, 0)$$

$$\text{Giving } M(\theta, \phi) = \begin{pmatrix} \cos\theta \cos\phi & -\sin\phi & \sin\theta \cos\phi \\ \cos\theta \sin\phi & \cos\phi & \sin\theta \sin\phi \\ -\sin\theta & 0 & \cos\theta \end{pmatrix}$$

For rotation matrices $M^{-1} = M^T$

$$M^{-1} = \begin{pmatrix} \cos\theta\cos\phi & \cos\theta\sin\phi & -\sin\theta \\ -\sin\phi & \cos\phi & 0 \\ \sin\theta\cos\phi & \sin\theta\sin\phi & \cos\theta \end{pmatrix}$$

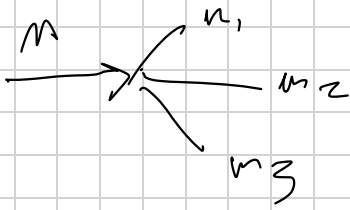
2) we have

$$T_{max} = \frac{2m_e c^2 \beta^2 \gamma^2}{1 + \frac{2r_{me}}{m} + \left(\frac{m}{m}\right)^2}$$

and $E_S = T_{max} \cos^2 \theta$

$$\theta = \cos^{-1} \sqrt{\frac{E_S}{T_{max}}}$$

3)



$$P_M = \sum_i P_i^M$$

define

$$P_{ij} = P_i + P_j$$

Have constant $M = \sum E_i$ Define $m_{23} = m_2 + m_3$

$$M = m_1 + m_{23}$$

Now 2-body decay

$$E_1 = \frac{M^2 + m_1^2 - m_{23}^2}{2M}$$

$$E_1 = \frac{M^2 + m_1^2 - (m_2 + m_3)^2}{2M}$$

generalized

$$E_i = \frac{M^2 + m_i^2 - (m_j + m_k)^2}{2M}$$

4) Event type is so important